```
  GNU nano 7.2                  combined_redirection.c *
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

int main(void) {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        int fd = open("combined_output.txt",
                      O_WRONLY | O_CREAT | O_TRUNC, 0644);

        if (fd < 0) {
            perror("open");
            _exit(1);
        }

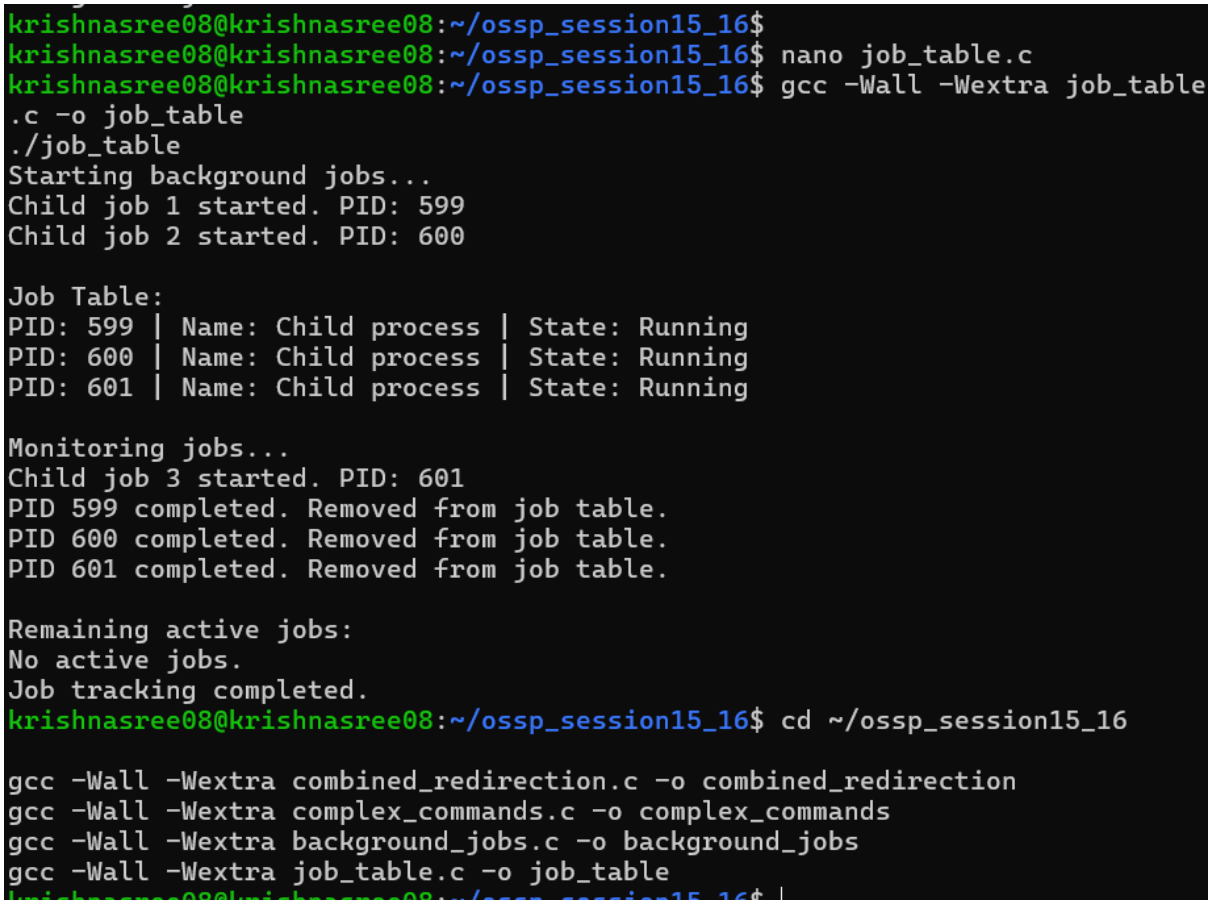
        if (dup2(fd, STDOUT_FILENO) < 0 ||
            dup2(fd, STDERR_FILENO) < 0) {
            perror("dup2");
            close(fd);
            _exit(1);
        }

        close(fd);
        execlp("ls", "ls", ".", "/folder_that_does_not_exist",
               (char *)NULL);
        perror("execlp");
        _exit(1);
    }

    waitpid(pid, NULL, 0);
```

```
krishnasree08@krishnasree08:~/ossp_session13_14$ mkdir -p ~/ossp_session15_1
6
cd ~/ossp_session15_16
krishnasree08@krishnasree08:~/ossp_session15_16$ nano combined_redirection.c
krishnasree08@krishnasree08:~/ossp_session15_16$ gcc -Wall -Wextra combined_
redirection.c -o combined_redirection
./combined_redirection
cat combined_output.txt
Combined redirection completed.
Check combined_output.txt for output and errors.
ls: cannot access '/folder_that_does_not_exist': No such file or directory
.:
combined_output.txt
combined_redirection
combined_redirection.c
krishnasree08@krishnasree08:~/ossp_session15_16$ nano complex_commands.c
krishnasree08@krishnasree08:~/ossp_session15_16$ ^C
krishnasree08@krishnasree08:~/ossp_session15_16$ gcc -Wall -Wextra complex_c
ommands.c -o complex_commands
./complex_commands
Complex command execution completed.
Result saved in output.txt:
2
krishnasree08@krishnasree08:~/ossp_session15_16$ nano background_jobs.c
krishnasree08@krishnasree08:~/ossp_session15_16$ gcc -Wall -Wextra backgroun
d_jobs.c -o background_jobs
./background_jobs
Parent process continues immediately.
Background job started. PID: 585
Background job PID: 585
You can continue using the terminal while the job runs.

Background job completed.
Background job finished with status 0.
```



Overall Result

Thus, the OSSP Sessions 15 and 16 programs demonstrate combined input/output redirection, complex command execution using pipes, background process management, job tracking, process synchronization, and resource management.